

1 Extended Application Configuration

1.1 Overview of INI configuration files

INI file	Configurations
ctaip.ini	Host name, terminal number, virtual keyboard on/off, inputs/outputs
ctaip.mld	Translation of labels
ctaipbut.ini	Configuration of buttons: order, positioning, icons, if necessary licenses
ctaisplay.ini	Configuration of grid layout; basic screen: height of tables and buttons; layout of BDE comments; OP info, machine info
dialog.ini	Configuration of font type/size in dialogs and of tab sizes in workflow dialogs
keyboard.ini	Configuration of the virtual keyboard: size and behavior

1.2 General

1.2.1 Identification of lists / elements in the terminal

The following shortcut activates information about available lists or elements in the terminal:

CTRL + ALT + F6

or in

AIP DEBUG menu: Further debug functions → Activate hints (scroll down)

A tooltip is shown when hovering the mouse pointer over a table or element.

The value "table" identifies the list:

Operation	Target qt
0010 Produce wheel cases	200
Name:AuftGrid Class:tMpdvExtendedGrid Source:c:\aip\spool\anr.lst Filter:MNR=00001021&AST_OPT_PKENN=L F Order: Table:Auftragsliste IniFile:c:\aip\ctaipay.ini Font:ARIAL , 11 , 1(DEFAULT_CHARSET)	

1.3 Modifications to ctaipbut.ini

1.3.1 General

The complete standard INI files are located on the server directory \mip\ctnet\win\aip2

Any deviations are developed in specific, customized directories.

e.g. for customized terminal groups \mip\<SystemNo>\custom\aip2\tgrp_xxx

(xxx = number of terminal group)

An empty file (ctaipbut.ini) is generated in these directories. All customized sections e.g. [ANR-ALL-Page1] are copied to this file. Then the configuration is performed in this file.

After restarting the terminal, files from the main directory \mip\ctnet\win\aip2 are merged with files from customized directories (e.g. \mip\<SystemNo>\custom\aip2\tgrp_xxx). The merged file is then transferred to the local terminal directory C:\MPDVAIP2.



The directory \mip\ctnet\win\aip2 must not be changed, otherwise AIP2 might no longer work properly. In addition, default files are stored there and any changes made will be lost after updating (e.g. service pack)!

1.3.2 Modifications to the toolbar

A ctaipbut.ini file including the modified section is stored in the customized terminal directory (e.g. if the toolbar is changed for terminal groups: \mip\<SystemNo>\custom\aip2\tgrp_xxx\).

Example:

The order of buttons should be changed in the "machines" section of the AIP2 basic screen (position button "change status" first and then the "lock production status" button).

Initial configuration

```
[MNR-ALL-Page1]
1=M_INFO,L,,InfoRed.png
; switch basic screen from list view to single machine view:
2=VIEW,L,,VirtualTourSmall.png
3=P_SPERRE,L,Produktionsstatus sperren,Security Risk.png
4=M_MST,L,Status ändern,Status Flag Yellow.png
5=A_AN*,R,Arbeitsgang anmelden,SyBluPly.png
```

New configuration

```
[MNR-ALL-Page1]
1=M_INFO,L,,InfoRed.png
; switch basic screen from list view to single machine view:
2=VIEW,L,,VirtualTourSmall.png
3=M_MST,L,Status ändern,Status Flag Yellow.png
4=P_SPERRE,L,Produktionsstatus sperren,Security Risk.png
5=A_AN*,R,Arbeitsgang anmelden,SyBluPly.png
```

1.3.3 Modifications to button labeling

A ctaipbut.ini file including the modified section is stored in the customized terminal directory (e.g. if the toolbar is changed for terminal groups: \mip\<SystemNo>\custom\aip2\tgrp_xxx\).

The required sections can just be inserted if a customized ctaipbut.ini file already exists, e.g. due to changes to the order of buttons.

Example:

Labeling of buttons in the "operation" section should be changed as follows:

- "Partial confirmation" --> "Part. conf."
- "Interrupt operation" --> "Interrupt OP"
- "Log off operation" --> "Log off OP"

Initial configuration

```
[ANR-ALL-Page1]
1=A_INFO,L,,InfoBlue.png
2=A_TR,R,Teilrückmeldung,SyBluAdd.png
3=A_UN*,R,Arbeitsgang unterbrechen,SyBluPau.png
4=A_AB*,R,Arbeitsgang abmelden,SyBluStp.png
```

New configuration

```
[ANR-ALL-Page1]
1=A_INFO,L,,InfoBlue.png
2=A_UN*,R,AG unterbrechen,SyBluPau.png
3=A_TR,R,Teiltrück.,SyBluAdd.png
4=A_AB*,R,AG abmelden,SyBluStp.png
```

1.3.4 Modifications to icons

General

Generate the file pict_cust.zip in the customer specific directory \mip\<SystemNo>\custom\.

Enter customer-specific icons (e.g. custom tools.png) to the file pict_cust.zip.

Note:

The file pic.zip contains all icons used at the terminal.

The file name for the button icon can be changed in the customized section of ctaipbut.ini.

Example:

Changing the icon for the button "Log on operation" from SyBluPly.png to Custom Tools.png.

Initial configuration

```
[MNR-ALL-Page1]
1=M_INFO,L,,InfoRed.png
; switch basic screen from list view to single machine view:
2=VIEW,L,,VirtualTourSmall.png
3=P_SPERRE,L,Produktionsstatus sperren,Security Risk.png
4=M_MST,L,Status ändern,Status Flag Yellow.png
5=A_AN*,R,Arbeitsgang anmelden,SyBluPly.png
```

New configuration

```
[MNR-ALL-Page1]
1=M_INFO,L,,InfoRed.png
; Switch basic display between list display and single machine display:
2=VIEW,L,,VirtualTourSmall.png3=P_SPERRE,L,Lock production status,Security Risk.png
4=M_MST,L,Change status,Status Flag Yellow.png
5=A_AN*,R,Logon operation,Custom Tools.png
```

1.4 Modifications to ctaip2.ini

1.4.1 General

The complete standard INI files are located on the server directory \mip\ctnet\win\aip2

Any deviations are developed in specific, customized directories.

e.g. for customized terminal groups \mip\<SystemNo>\custom\aip2\tgrp_xxx

(xxx = number of terminal group)

An empty file (ctaip2.ini) is generated in these directories. All customized sections e.g. [ANR-ALL-Page1] are copied to this file. Then the configuration is performed in this file.

After restarting the terminal, files from the main directory \mip\ctnet\win\aip2 are merged with files from customized directories (e.g. \mip\<SystemNo>\custom\aip2\tgrp_xxx). The merged file is then transferred to the local terminal directory C:\MPDV\AIP2.



The directory \mip\ctnet\win\aip2 must not be changed, otherwise AIP2 might no longer work properly. In addition, default files are stored there and any changes made will be lost after updating (e.g. service pack)!

1.4.2 Enter user fields in a table

Overview

A ctaip2.ini file including the modified section is stored in the customized terminal directory (e.g. if the toolbar is changed for terminal groups: \mip\<SystemNo>\custom\aip2\tgrp_xxx\).

ctaip2.ini is configured in two steps:

- Activate the additional loading of the user fields for operation- or order-related XML files in the section [Custom Userfields ANR]. Activate the user fields for machines in the [Custom Userfields MNR] section.
- Configure the field to be displayed in the grid in the section (e.g. to display the additional field in the order list), e.g. [Order list].

Below both steps are explained in detail.



If the user field is only to be displayed in the GUI with XML configuration, the configuration for display in the grid in the [Order List] or [Machine List] can be omitted.

Available user field in machine and order lists

All identifiers of user fields and also other fields that can be reloaded are located in the headers.dat file in the "spool" directory of the terminal. It consists of four lines:

Start of the row	Content
10 ...	Machine list: Fields that are always included in the list.
*10 ...	Machine list: Fields that can be reloaded.
11 ...	Order list: Fields that are always included.
*11 ...	Order list: Fields that can be reloaded.

The following user fields can be reloaded:

- Machine list:
 - FU:1 to FU:66:
Machine user field
- Order list:
 - ANR_FU_1 to ANR_FU_66:
Operation user fields
 - AUNR_FU_1 to AUNR_FU_66:
Order user fields
 - MNR_FU_1 to MNR_FU_66:
Machine user fields
 - VERARBCODE_FU_1 to VERARBCODE_FU_66:
Processing code user fields
 - AGR_FU_1 to AGR_FU_66:
Operation status user fields (reserved for customizations)

Example 1: User fields in the operation list

User field 1 of the operation should be entered in the order list with the name " Order date ".

User field 66 of the machine with the name "My long user field" should be added to the order list.

Step 1 ➔ Field definition of the section [Custom Userfields ANR]

```
[ Custom Userfields ANR ]
GRID_LIST_TYP=ANR
; Additional fields in list of operations
ANR_FU_1= ; User field 1 of operation, MyDate FU:1 [operations list]
MNR_FU_66= ; User field 66 of machine, My long user field [operations list]
```

Step 2 ➔ add the field to the grid [order list]



If the user field is only to be displayed in the GUI with XML configuration, the configuration in the [Order List] can be omitted.

```
[Order list]
GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite
GRID_LIST_TYP=ANR
EXAMINE_BITMAP1=B1,OPT_INFOAN,T=Attach Notes.png
EXAMINE_BITMAP2=B2,OPT_INFOAI,T=Text Document.png
ATK=C25,100,L,Article
ANR_FU_1=dd.mm.yyyy,90,L,MyDate FU:1
MNR_FU_66=C40,150,L,My long user field of machine
```

Example 2: User field in the machine list

User field 66 of the machine with the name "My long user field" should be added to the machine list.

Step 1 ➔ Field definition of the section [Custom Userfields MNR]

```
[ Custom Userfields MNR ]
GRID_LIST_TYP=MNR
; Additional fields in list of machines
ANR_FU_66= ; User field 66 of machine, My long user field
```

Step 2 ➔ Enter the field for the grid [Machine list]



If the user field is only to be displayed in the GUI with XML configuration, the configuration in the [Machine List] can be omitted.

```
[Machine list]
GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite
GRID_LIST_TYP=ANR
EXAMINE_BITMAP1=B1,OPT_INFOAN,T=Attach Notes.png
EXAMINE_BITMAP2=B2,OPT_INFOAI,T=Text Document.png
ATK=C25,100,L,Article
FU:66=C40,150,L,My long user field
```

Explanation of the syntax using the order list

ANR_FU_1 = user field of of the operation

dd.mm.yyyy = formatted as date

90 = number of pixel for the column

L = aligned to the left

MyDate FU:1 = column header

MNR_FU_66 = user field 66 of the machine

C40 = alphanumeric field with 40 digits

150 = number of pixel for the column

L = left aligned

My long user field = column header

1.4.3 Change order of columns in AIP2

A ctaip2.ini file including the modified section is stored in the customized terminal directory (e.g. if the toolbar is changed for terminal groups: \mip\<SystemNo>\custom\aip2\tgrp_xxx\).

The required sections can just be inserted if a customized ctaip2.ini file already exists, e.g. due to changes to the order of buttons.

Example:

The "order" column should be displayed in the first place and then the "article" column.

Initial configuration

```
[Order list]
```

```
...
```

```
ATK=C25,100,L,Artikel
```

```
AUNR=C10,85,L,Auftrag
```

```
ANR_FU_65=C30,150,L,Artikelbezeichnung 2
```

```
AGNR=C4,39,R," "
```

New configuration

[Order

list]

...

AUNR=C10,85,L,Auftrag

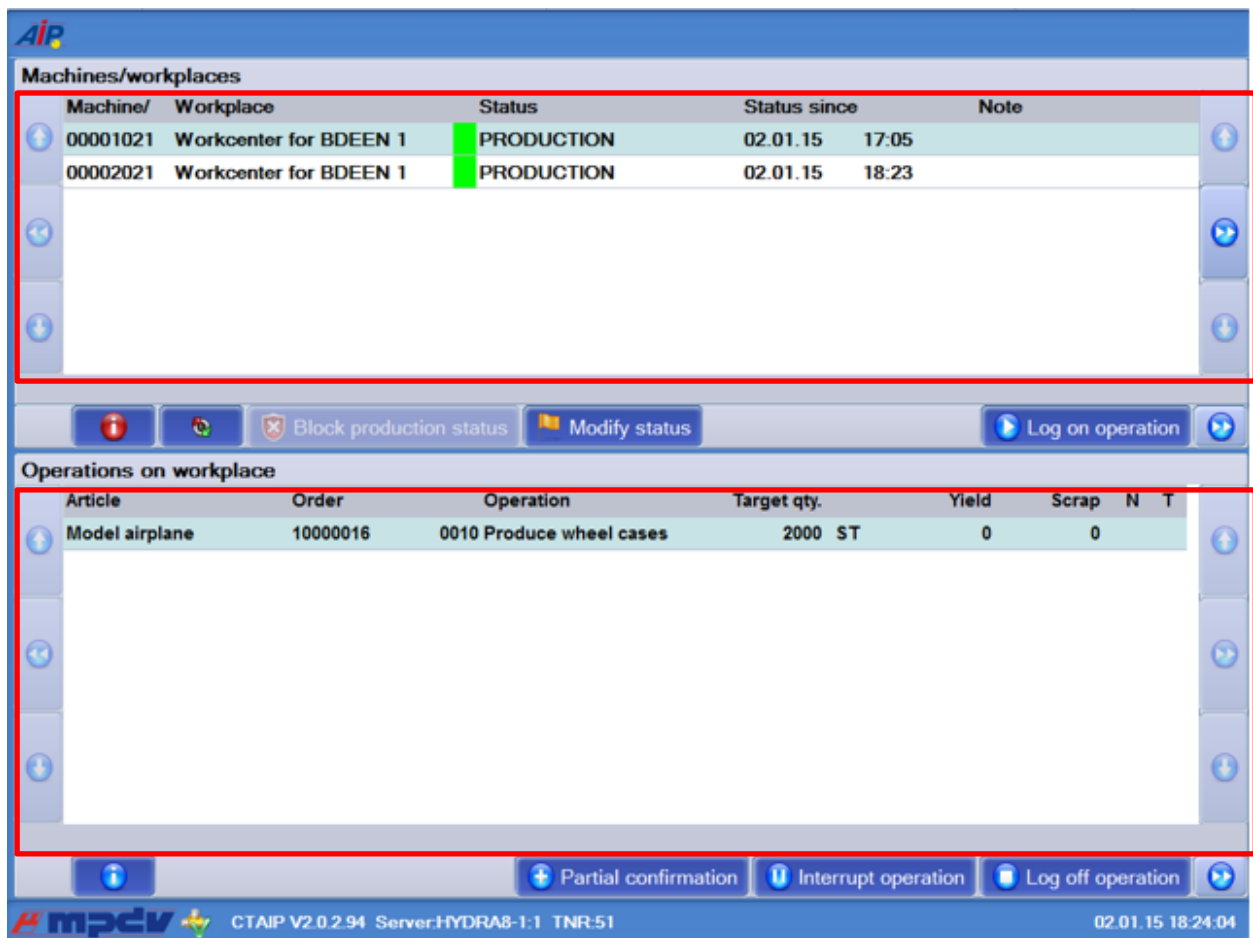
ATK=C25,100,L,Artikel

ANR_FU_65=C30,150,L,Artikelbezeichnung2

AGNR=C4,39,R," ..

1.4.4 Changing the height of AIP2 lists

Changing the height of lists (operation and machine list) in the basic screen [MainView1] of AIP2.



The configured heights are scaled to the current height. Consequently, the total sum of entered heights is irrelevant.

The height of lists and elements of the basic screen (machines, order grid, 3rd list, toolbar) can be configured in section [MainView1] of the customized ctaipplay.ini file.

```
[MainView1]
; Values are scaled to match the current resolution
; in order to use the full screen. Percentage values can also be entered
OrderGridHeight=440
MachineGridHeight=380
;List3GridHeight=200
ButtonBarHeight=50
```

Explanation: SYNTAX of order list

OrderGridHeight = height of the order list (indicated in pixels)

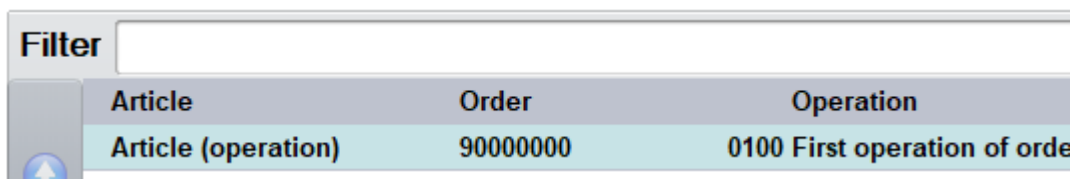
MachineGridHeight = height of the machine list (indicated in pixels)

List3GridHeight= height of the third list (tools, staff, batches,...) (indicated in pixels)

ButtonBarHeight= height of the toolbar (indicated in pixels)

1.4.5 Changing the filter function in tables

The filter function is activated for many tables. The filter function You can configure on which column the filtering is to take effect.



Requirements:

- The field for filtering is activated for the table in the dialog configuration (field attribute "AUTOFILTERFIELD").
- There is an entry "AUTOFILTERCOL" in the configuration file ctaipplay.ini which specifies the column to be filtered.

Proceed as follows:

1. Use the shortcut "Ctrl+Alt+F6" in the AIP2 to activate the tooltip.
2. Display the required table on the AIP that already has a filter field.
3. Use the tooltip to identify the affected section in the ctaipplay.ini file by moving the mouse pointer over the desired table. E.g.,... Cfg: **WF@AGNR ...**.
4. Find the section in the standard configuration file <server>\mip\ctnet\win\aip2\ctaipplay.ini and copy the section to your customized global file <server>\mip\<SystemNo>\custom\aip2\ctaipplay.ini or to a terminal group specific file <server>\mip\<SystemNo>\custom\aip2\tgrp_<TerminalGrpup>\ctaipplay.ini

5. Change the value of the existing AUTOFILTERCOL property in the copied customer-specific section. You can find information on available columns in other attributes of the section or in the file headers.dat. If the property AUTOFILTERCOL is not available, the filtering cannot be changed by configuration. If the filtering is not activated, you first need to check further requirements and activate the filtering via a customizing.



The setting AUTOFILTERCOL=<ALL> ensures that the filter value applies to all columns of the table. Following, a row is displayed if the column contains filter text.

Example<server>\mip\<SystemNo>\custom\aip2\ctaiplay.ini:

```
*****
;
; ctaipplay.ini (customer's)
;
; -----
[WF@ANR]
CMD=DLG=LIST;11|MOD=V|MNR=<MNR>|
MODE=CMD:MODE=#LOCKED#|DATALOCKUNTILSHOW=TRUE|ADDCALCULATEDFIELDS=A|FOCUSLISTITEM=ANR=<ANR>|
FILTER=
SECTION=Sequencing List (Auto)
DATAFIELDS=ANR & AGNR & ATK & ATKBEZ & AGBEZ & *CHPFL=CHPFL & ANR.ATK=ATK & ANR.ATKBEZ=ATKBEZ & ANR.AGBEZ=AGBEZ &
ANR.SGR:GUTP=SGR:GUTP & ANR.EGR:GUTP=EGR:GUTP & ANR.EGR:AUSP=EGR:AUSP & ANR.AUNR=AUNR & RMNR=ANR_RMNR & ANR.FERTIG=FERTIG
FILE=vlist.<MNR>.lst
AUTOFILTERCOL=AGNR
*****
```

1.4.6 Cyclic reload of the sequencing list

The sequencing list is cyclically updated on the AIP. The setting of the cycle takes place in ctaipnet.ini
mdereoadvorgabeliste=600

If the sequencing list is to be reloaded each time the operation logon dialog is called, the parameter CMD:MODE=#LOCKED#| must be removed from the section [WF@ANR] in the MODE entry of the ctaipplay.ini. If you want to make a custom implementation, you should copy this section into a new (empty, if not already existing) file ctaipplay.ini and delete the above entry.

1.5 Changes to ctaip.ini

1.5.1 General

The file ctaip.ini is not merged with a file stored in the server. The file must be edited locally in the terminal.

1.5.2 Start Third-Party Application from AIP

Starting a third-party application is configured as follows:

- Configure a button in the ctaipbut.ini file

- Configuration of the function

AIP allows for the integration of buttons starting third-party applications in all toolbars. These buttons

- start third-party applications provided they are not running
- bring third-party applications to the front when they are running

Configuration of buttons

The first button starting a third-party software is configured as "USER1" in ctaipbut.ini.

Further buttons starting third-party software can be configured as "USER2" to "USER9" in the ctaipbut.ini file.

Example:

Configuration of a new button. The button is to be displayed for a specific terminal group in the "machines" section of the AIP2 basic screen. It is configured in the ctaipbut.ini file specific to terminal groups in the server.

\\mpip\<SystemNo>\custom\aip2\tgrp_xxx\ctaipbut.ini)

```
[MNR-ALL-Page1]
1=M_INFO,L,,InfoRed.png
; Switching the basic screen between list view and presentation of individual machines:
2=VIEW,L,,VirtualTourSmall.png
3=P_SPERRE,L,Produktionsstatus sperren,Security Risk.png
4=M_MST,L,Status ändern,Status Flag Yellow.png
5=A_AN*,R,Arbeitsgang anmelden,SyBluPly.png
6=USER1,R,Notepad
```

Configuration of the function

The function is configured in section [ext. software] of the local terminal configuration file "ctaip.ini":

Example:

```
[ext. software]
Button=Notepad
WindowName=Notepad
ProgFileName=C:\Program Files (x86)\Notepad++\notepad++.exe
SearchParts=On
```

Please note:

"SearchParts=" → If this entry is set, it is sufficient to enter the program name only partly in WindowName.

1.5.3 Remember staff badge number

The person memorized by the terminal only changes if the person logs on with the order. (A_P_AN instead of A_AN)

The memorized person is removed from the memory when the person explicitly logs off from the machine (the same applies for "log off all").

This pre-assignment can be suppressed. Configure "default=0" and set the field attribute SETVALUE in the dialog configuration.

The number is pre-assigned in all dialogs for order postings, status changes and when batches are posted C_UMB, C_GEN, CA_WL.

The memorized persons are deleted in the memory for all machines when shifts change or at the beginning of a new shift (**Note:** If the shift changes at one machine of the terminal, the persons at the other machines of the terminal are also deleted)!

This can be configured via the entry "HoldPersonInfo=on" in section [SYSTEM] of the ctaip.ini file.

Example:

```
[System]
.....
HoldPersonInfo=on
```

1.6 Hide virtual keyboard

The virtual keyboard can also be switched off if the terminal is connected to a real keyboard. This can be configured in section [SYSTEM] of the local ctaip.ini file. Example:

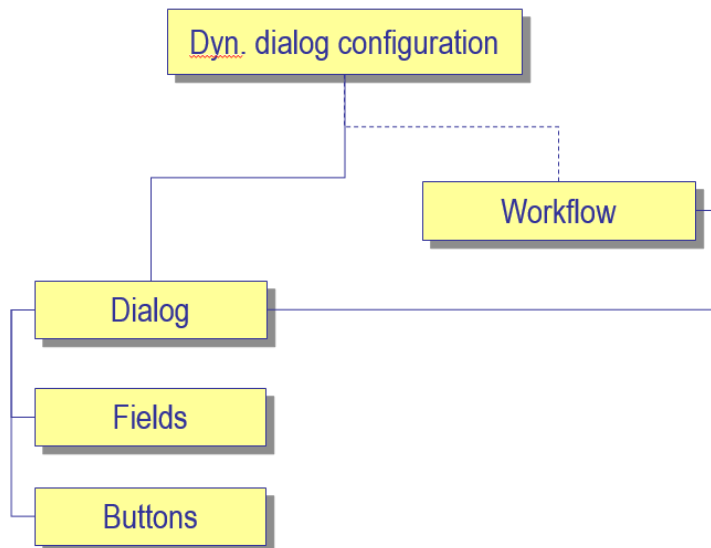
```
[SYSTEM]
Parameters=-t
```

Syntax:

+t/-t --> enables/disables the virtual keyboard; irrespective of the terminal type

1.7 Dynamic dialogs

1.7.1 Overview



1.7.2 AIP2 dialog types

AIP2 provides the following dialog types:

- AIPDEF – Default dialogs (customization)
- AIPTGRP – Dialogs for specific terminal groups (configuration)
- AIPTNR – Dialogs for specific terminals (customization)



You can only create/change dialogs for specific terminal groups. You can change existing dialogs for a specific terminal.

But default dialogs **cannot** be changed.

The terminal has to be rebooted after dialogs were changed.

1.7.3 Dialogs for specific terminal groups

You can make configurations for specified terminal groups.

Before starting the configuration, make a backup copy of the concerned dialogs in a backup group (e.g. AIPTGRP 999). (In case old backups exist, delete them beforehand).

Activate the new dialogs and reboot the terminal.

Example:

Special posting dialogs should be used for terminal group xxx. The workflows and dynamic dialogs are assigned to this terminal group.

How to proceed

1. Assign terminal to terminal group
2. Copy workflows to terminal group xxx
3. Copy dynamic dialogs to terminal group xxx
4. Activate dialogs

Assign terminal (MOC)

Menu: System administration --> Terminals --> Terminal groups

Copy workflows (MOC)

Menu: System administration --> Terminals --> Workflow

Copy complete workflow configuration from AIPDEF 0 to AIPTGRP xxx.

Copy dynamic dialogs (MOC)

Menu: System administration --> Terminals --> Dynamic dialogs

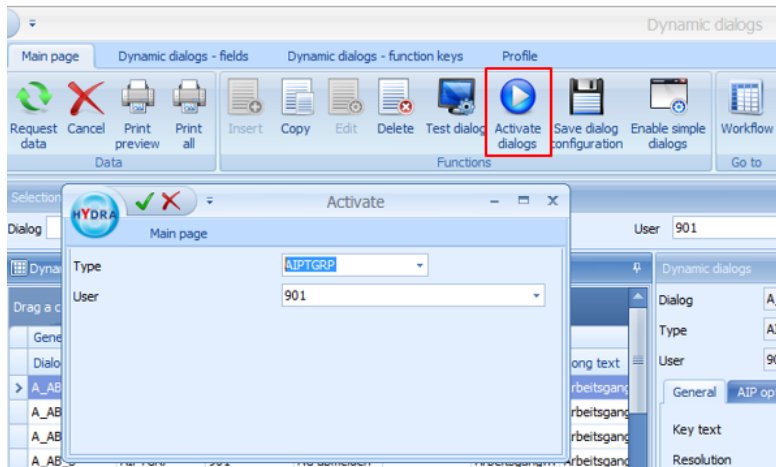
Copy the complete dialog configuration from AIPDEF 0 to AIPTGRP xxx

Activate dialogs (MOC)

Menu: System administration --> Terminals --> Dynamic dialogs

Button "Activate dialogs"

Dialog input: Type =AIPGRP ; User=xxx



Note:

You can delete all dialogs of a terminal group if you select all rows of the terminal group (AAPTGRP).

1.7.4 Hide fields (for specific terminal groups)



Identify the dialogs used on the AIP2 terminal.

Using the shortcut Ctrl + ALT + F6, a tooltip indicating the dialog name is shown.

General procedure:

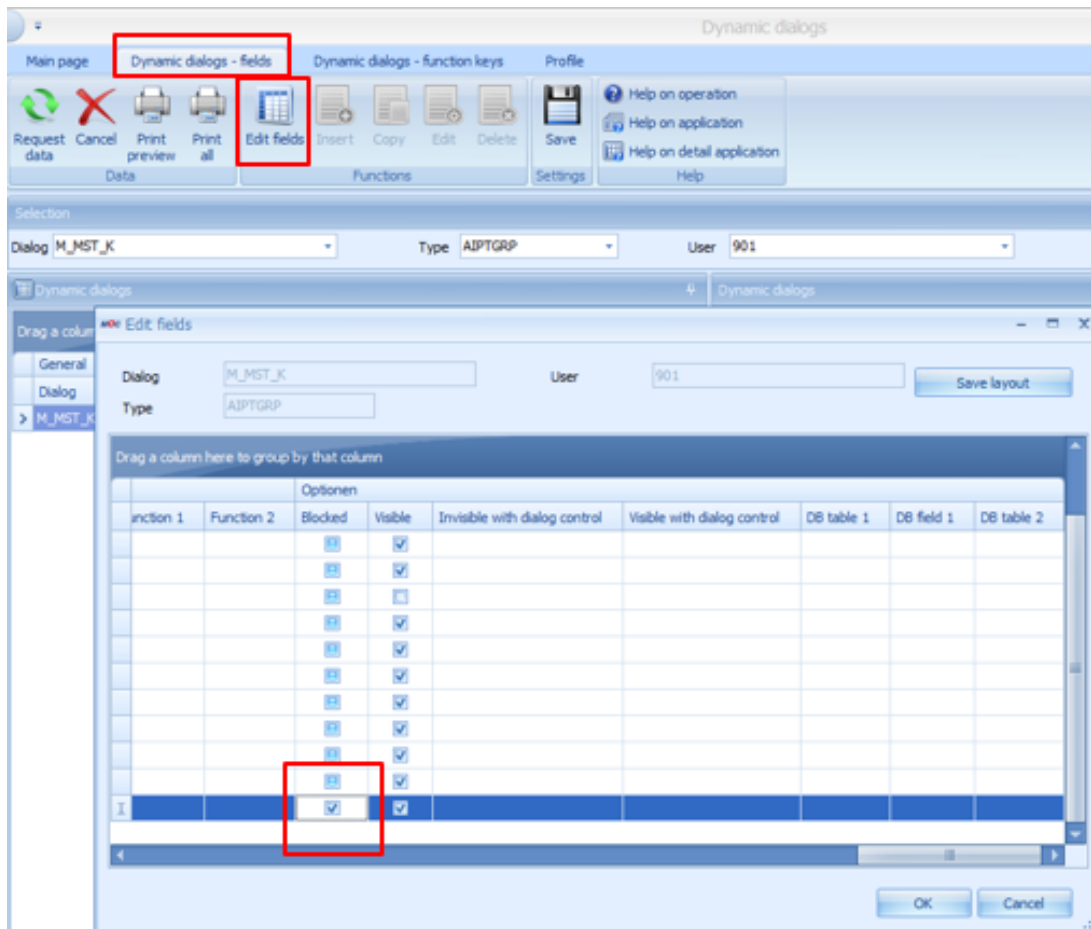
- Identify the dialog where a field should be hidden
- Change and activate dynamic dialogs of terminal group xxx.

Edit dynamic dialogs (MOC)

Menu: System administration --> Terminals --> Dynamic dialogs

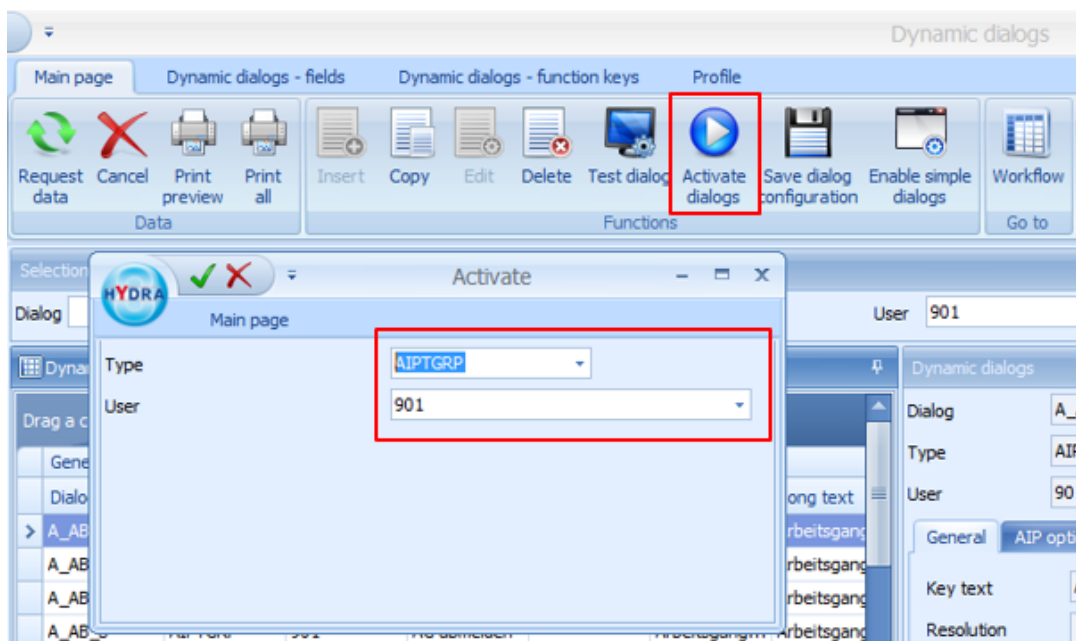
Select the dialog for a specific terminal group and start the edit mode via the menu tab "dynamic dialogs - fields" and the button "edit fields"

Choose the required field and check the option "blocked"



Activate dialog (MOC)

Activate dialogs for specific terminal groups



1.7.5 Default assignment in dialog fields (for specific terminal groups)

General procedure:

- Identify the dialog where a field should be completed with default values
- Change and activate dynamic dialogs of terminal group xxx.

Edit dynamic dialogs (MOC)

Menu: System administration --> Terminals --> Dynamic dialogs

Select the dialog for a specific terminal group and start the edit mode via the menu tab "dynamic dialogs - fields" and the button "edit fields"

Set field "field attribute 2" to "SETVALUE"

Add field "default"

Allowed characters for dynamic dialog fields

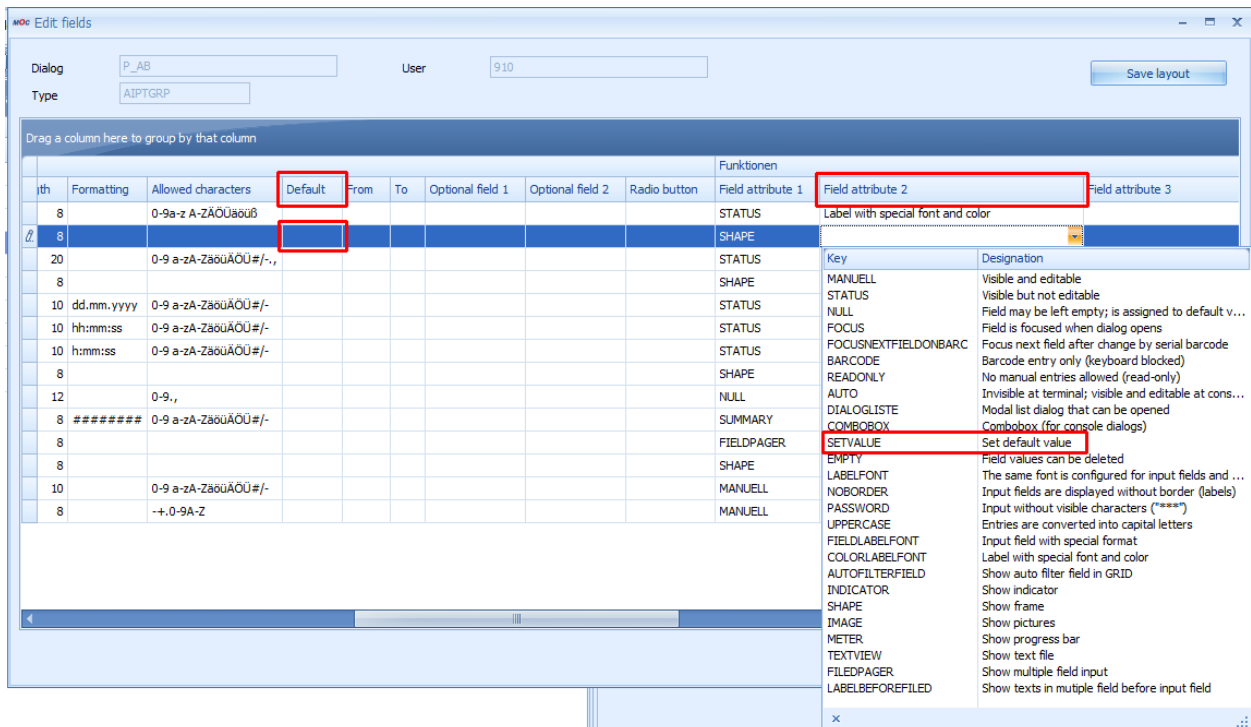
The minus character "-" must always be placed at the end to prevent it from being mistaken for the character used for the definition of "from" - "to" ranges.



Example:

a-z A-Z/,-. is interpreted as range from a to z and A to Z but in this case also as range from "/" to ","

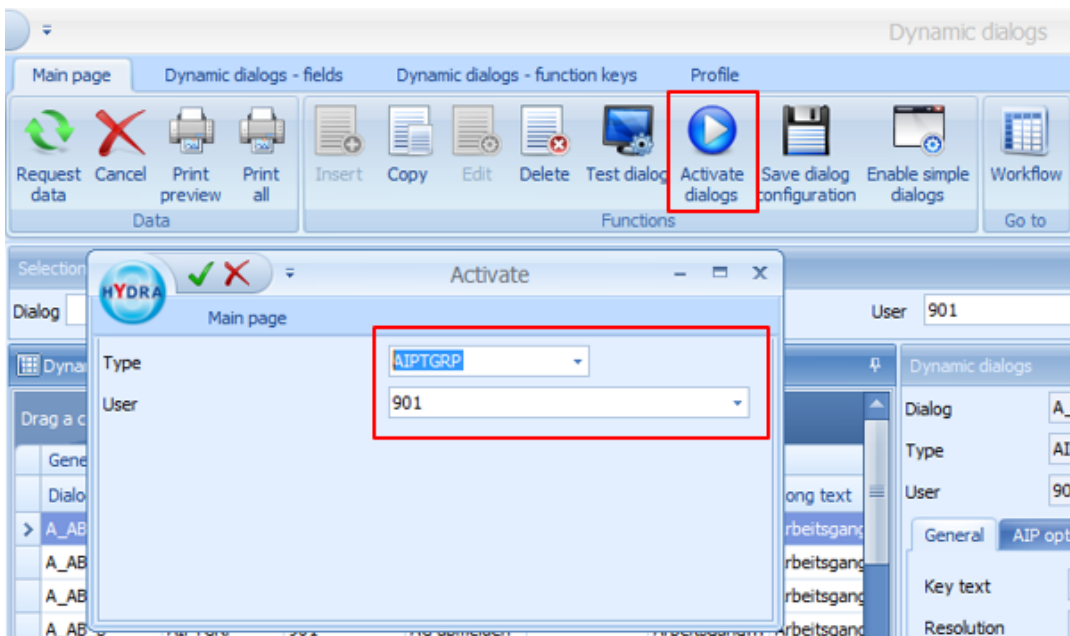
a-z A-Z/,-. is interpreted as range from a to z and A to Z and as the allowed characters / , . and -



Activate dialog (MOC)

Activate dialogs for specific terminal groups

Change field name (for a specific terminal group)



General procedure:

- Identify the dialog where a field name should be changed

- Change and activate dynamic dialogs of terminal group xxx.

Edit dynamic dialogs (MOC)

Menu: System administration --> Terminals --> Dynamic dialogs

Select the dialog for a specific terminal group and start the edit mode via the menu tab "dynamic dialogs - fields" and the button "edit fields"

Change field contents of the column "text".

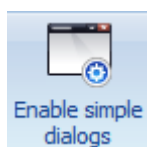
Field no.	Text	Unit	HYD
1	Operation		
2	Artide		
3	Artide Designation		
4	Comment, important		
5	Comment 2		
6	Plan. duration		
7	Coll...		

Activate dialog (MOC)

Activate dialogs for specific terminal groups

1.7.6 Activate simplified dialogs

There are simplified dialogs for: logging on operations, reporting partial quantities for operations, interrupting operations, logging off operations.



Use the button Enable simple dialogs to store the simplified dialogs for the standard AIPDEF 0 in the workflow.

How to proceed:

- Menu: System administration --> Terminals --> Dynamic dialogs --> Button "Enable simple dialogs"
- Enable dialogs for AIPDEF 0

Only one dialog is entered in the workflow if simple dialogs are in use.



Once simplified dialogs have been activated, it **cannot** be undone by way of configuration. This can only be changed by customizing the system, which has to be ordered from MPDV.

Activation via the standard dialogs AIPDEF 0.

1.8 Customizing files

1.8.1 Terminal script files

File names and directories of default/customized terminal scripts must be named as follows.

	AIP 2	Description
MPDV	.lctnet\winlaip2\etc\laip_mpdv.zip	MPDV standard (not used)
MPDV	.lctnet\winlaip2\etc\mpdv-aip.zip	MPDV standard
CUST	.lcustom\userexit\laip2_<customer number>.zip	Customization with customer number
CUST	.lcustom\userexit\laip2_<project>.zip	Customization with project abbreviation

“aip_” is added as prefix to terminal script files for AIP2.

	PRIOR	AIP 2	Description
MPDV	1	.laip_system_mpdv.scr .laip_<dialog>_mpdv.scr	MPDV standard (not used)
MPDV	2	.laip_mpdv-system.scr .laip_mpdv-<dialog>.scr	MPDV standard
CUST	1	.laip_system_<customer no.>.scr .laip_<dialog>_<customer no.>.scr	Customization with customer number
CUST	2	.laip_system_<project>.scr .laip_<dialog>_<project>.scr	Customization with project abbreviation

ZIP files are only unpacked in live operation, once they have been successfully DOWNLOADED from the server. In DEMO mode there is no unpacking of terminal script ZIP files.